



Negative impact of self-reported executive problems in patients with functional/dissociative seizures: Results from a prospective long-term observational study

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ABSTRACT

Objective: Functional/dissociative seizures (FDS) are common and pose a considerable burden on both individual patients and healthcare systems. Cognitive complaints are frequent in patients with FDS. Previous studies on cognitive function in patients with FDS have yielded mixed results. We investigated executive function and its impact on long-term outcome in a prospective sample of patients with FDS.

Methods: Thirty-three inpatients (age range: 16–62 years) who had been admitted to the National Centre for Epilepsy in Norway with FDS underwent neuropsychological assessment for both tested and self-reported executive function, using the Behavior Rating Inventory of Executive Function, at baseline. Participants were evaluated for their medical status at a mean of 5.5 years (SD 2.4, range 1.9–10.9 years) after inclusion. To determine potential predictors of achieving cessation of FDS by time of follow-up a multivariate logistic regression analysis was conducted.

Results: At follow-up, 14/33 (42 %) of the participants were FDS-free. Self-reported, but not tested, executive dysfunction was an independent risk factor for ongoing FDS at follow-up.

Conclusion: Our findings indicate an association between greater self-reported executive problems and poorer long term outcomes in patients with FDS. Screening for executive dysfunction with a self-report questionnaire may yield relevant information that could be used to guide psychological interventions.

1. Introduction

Functional dissociative seizures (FDS), also referred to as psychogenic nonepileptic seizures, are considered a functional neurological disorder (FND). FDS are characterized as superficially resembling epileptic seizures, but are not associated with epileptiform activity [1]. FDS are commonly encountered by neurologists, sometimes misdiagnosed as epilepsy resulting in treatment with antiseizure medication (ASM) and a missed opportunity for treatment of the functional disorder. FDS pose a substantial burden on the patients, their families, and healthcare systems [2]. The incidence of FDS diagnoses is estimated to be around 3/100 000 per year, while the prevalence is approximately

24/100 000 [3]. FDS are commonly associated with other FNDs, and psychiatric disorders in general, such as post-traumatic stress disorders, personality disorders, depression and anxiety [4]. The prognosis for improvement is often poor, with around two-thirds of adults with FDS continuing having seizures several years after diagnosis [5,6]. While psychotherapeutic interventions are the mainstay of treatment for FDS [7], drop out from follow-up and treatment is common [8].

Cognitive complaints are frequent in patients with FDS and have a substantial impact on their health-related quality of life [9]. Studies of cognitive function in patients with FDS have yielded mixed results. When cognitive impairment is reported, it often shows a general dysfunction affecting multiple domains [10], with alterations in

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attention and executive function (EF) being the most consistent findings [11]. EFs are a group of top-down mental processes, linked to the pre-frontal cortex, that help us concentrate, plan, take action, and regulate our emotions. EF is crucial for goal-directed behavior and involves the ability to control and regulate cognition and behavior, and to adapt to changing situations [12]. Consequently, EFs are skills essential for mental and physical health and adaptive social functioning. EF may thus play an important role in, for instance, engagement with, and adherence to, treatment plans for patients with FDS.

In sum, although there are some indications of difficulties with EF in patients with FDS, little is known if alterations in cognitive function, in general, and especially in EF, are related to long-term outcome with regards to seizure freedom and quality of life.

In the present study we investigated the relationship between EFs and seizure freedom at long-term follow up in a prospective sample of patients with FDS.

2. Methods

2.1. Participants

The selection of participants and study procedures have been described in detail elsewhere [13]. Briefly, patients with documented FDS (according to the definition from the International League Against Epilepsy; ILAE) [14] aged 16 years and older were prospectively included during their hospital stay at the National Centre for Epilepsy in Norway between 2009 and 2017. The patients were interviewed, and medical records were reviewed to ascertain the FDS diagnosis and obtain clinical characteristics, including medical and seizure history. Patients with comorbid epilepsy were excluded. Other exclusion criteria were: estimated IQ <70 and patients with such severe comorbid medical and/or psychiatric conditions that they were considered unsuitable for undergoing the planned assessments.

Sixty-two patients were invited to join the study, of whom four declined. Two patients did not meet our study criteria and were excluded. One patient died during the study and was excluded as undiagnosed concomitant epilepsy was considered possible. An additional two patients withdrew their consent during the study. Thirty-six participants underwent neuropsychological assessment at baseline. Of these, 35 provided follow-up data. Two patients were excluded due to IQ <70, thus the final sample consists of 33 patients.

Outcome data on seizure frequency were obtained via telephone interview by the first and last author (AV, MIL) at a mean of 5.5 years (standard deviation (SD) 2.4, range 1.9–10.9) and a median of 5.2 years after inclusion. The telephone interview contained questions about current seizure frequency, last seizure, actual educational and employment status, received treatment and the current self-rated health related quality of life.

2.2. Ethics

Informed written consent was obtained from the participants, or from both participants and parents for those under the age of 18 years. The study was approved by the Regional Committee for Medical Research Ethics, South-East Norway (2009/1078/REC South-East).

2.3. Neuropsychological assessment

The intelligence quotient (IQ) was estimated using the vocabulary and the matrix reasoning subtests from the Wechsler Abbreviated Scale of Intelligence (WASI), which is considered to provide a brief, reliable measure of cognitive ability for use in clinical and research settings [15]. Commonly used tests were employed to assess different aspects of EFs. Cognitive flexibility was assessed by the Trail-Making test B from the Halstead-Reitan Neuropsychological Battery (HRNB) [16]. Auditory working memory was evaluated by the Number-Letter Switching task

from WAIS-IV [17] and selective attention/response inhibition was assessed by the Color-Word Interference tests (CWIT) from the Delis-Kaplan Executive function (D-KEFS) test battery [18]. Attention span was measured by using the Digit span (forwards and backwards). Processing speed was evaluated by the Digit symbol subtests from WAIS-IV, as well as the Trail Making Test-A (REF).

All raw scores were converted to T-scores. We chose a T-score of 35 or below (or a percentile of 6.68 or below) equivalent to 1.5 SD below normative mean to represent the cutoff between normal and abnormal neuropsychological function. This is a commonly used impairment cutoff in neuropsychology [19,20]. A composite Executive Score was computed based on average T-scores for three executive subtests.

2.4. Behavioral evaluation

To assess behavioral and cognitive aspects of EF, we used the Behavior Rating Inventory of Executive Function – Adult version (BRIEF-A) [21] for participants aged 18 years or above and the Behavior Rating Inventory of Executive Function - Self Report Version (BRIEF-SR) for adolescents [22] for participants younger than 18 years. The standardized ratings scales have been developed to assess difficulties with executive functions in everyday life BRIEF consists of nine subscales, as well as three embedded validity scales assessing negative, atypical, and inconsistent response biases. Each question is rated on a 3-point Likert-type scale, “never”, “sometimes”, and “often”. The rating scales are summed into two major indexes: the Behavioral Regulation Index (BRI) and the Metacognition Index (MI), and these combined form the overall summary, the Global Executive Composite (GEC). The BRI is composed of the following four subscales: Inhibit, Shift, Emotional Control, and Self-Monitor. It is designed to measure the ability to maintain appropriate regulatory behavioral control and emotional responses. Appropriate behavioral regulation is thought to be a likely precursor to efficient metacognitive problem solving. The MI includes the following five subscales: Initiate, Working Memory, Plan/Organize, Task Monitor, and Organization of Materials (the BRIEF-SR does not generate the Initiate subscale, but includes the Task Completion subscale). In general, the MI reflects the ability to initiate activity, to sustain working memory, to plan and organize problem-solving approaches in a variety of contexts, and to organize belongings. The BRIEF-A has been found to be sensitive to the presence of executive dysfunction in a variety of conditions such as mild cognitive impairment [23], traumatic brain injury [24], and attention-deficit/hyperactivity disorder [25]. As BRIEF assesses difficulties with EFs, a higher T score is related to a higher degree of difficulties, as opposed to the neuropsychological test results where a higher T score reflects better function. For BRIEF results a T-score ≥ 65 (≥ 1.5 SDs) is used as the clinical cutoff in each domain, according to recommendations in the test manual. Scores above cutoff indicate significant difficulties with executive functions in everyday life.

2.5. Statistical analysis

Group comparisons (e.g., between patients who were seizure free at follow-up, defined as absence of FDS during the previous year, and those continuing to experience FDS at follow-up) were tested using Student's t-test for continuous and normally distributed variables, Mann–Whitney U-tests for ordinal or skewed variables, and chi-square tests or Fisher's exact tests for categorical data. All hypothesis testing was two-tailed. P values of <0.05 were considered statistically significant. To determine potential predictors of achieving cessation of FDS by time of follow-up, we performed multivariate logistic regression analysis. All statistical analyses were performed using Statistical Package for the Social Sciences (SPSS, version 26, IBM). Missing items were replaced with the mean of the answered items in the subscale, provided that at least half of that subscale had been answered.

3. Results

3.1. Baseline

Demographic and clinical baseline characteristics are provided in Table 1.

Table 2 shows the cognitive and behavioral findings at baseline. Two participants had a full-scale IQ below 70 and were therefore excluded, leaving 33 for the final analyses. Among the participants, 6.1 % scored below cutoff on the full-scale IQ and 21.2 % were below cutoff on the executive score. Concerning behavioral measures for EF, 27.8 % scored above cutoff on the BRI and 33.3 % scored above cutoff on the MI of the BRIEF.

3.2. Follow up

At follow-up, 14/33 (42 %) of the participants were seizure-free, whereas 28/33 (85 %) reported a seizure-reduction of ≥ 50 %. Table 3 shows group differences between patients who were seizure-free ($n = 14$) or not ($n = 19$) at follow-up with regards to demographics, clinical information, and neuropsychological and self-report measures of EFs. Patients who continued to have FDS at follow-up, had reported executive problems above the cutoff level more often, as assessed by the MI ($p = 0.02$). For IQ measures and tested EF, we found no significant group-level differences between patients who had become FDS-free and those who had not.

Table 4 shows that patients scoring above (indicating difficulties with executive function) or below cutoff on the MI Score were similar on all the baseline demographic and neuropsychological-test variables, except for a prior history of anxiety or depression ($p = 0.01$) and self-reported health-related quality of life scores ($p = 0.05$). An MI score above cutoff was associated with a BRI score above cutoff. With regards to outcome, a normal MI score was associated with FDS reduction ($p = 0.005$) and FDS freedom ($p = 0.02$) at follow-up.

We used logistic regression analysis to identify independent factors associated with an outcome of being FDS-free. Our small sample size limited the number of potential predictors that we could include. We therefore only included the MI and age at FS onset in the model. MI above cutoff was a predictor for ongoing FDS at follow-up, with an odds ratio (OR) of 10.47 (CI: 1.04–106.02, $p = 0.047$), see Table 5.

4. Discussion

The main finding in our study is that self-reported executive

Table 1
Baseline characteristics of the FDS-sample.

	N	Values
Female sex, n (%)	33	28 (84.8)
Age at presentation in years, mean (SD, range)	33	32.9 (13.8, 16–62)
Age at FDS onset in years, mean (SD, range)	32	26.4 (12.7, 8–56)
Diagnostic delay in years, mean (SD, range)	33	5.3 (7.7, 0.2–31.0)
FDS frequency per month, mean (SD, range)	33	13.6 (16.8, 0.2–70)
Education in years, mean (SD, range)	30	13.1 (2.9, 7–22)
Marital status, n (%)	33	
Married/partner		17 (51.5)
Single/separated		16 (48.5)
Employment, n (%)	30	
Employed/student		14 (46.7)
Unemployed		16 (53.3)
Psychiatric history for anxiety or depression, n (%)	30	15 (50)
Psychotherapy prior to inclusion, n (%)	31	19 (61.3)
HQoL-VAS, mean (SD, range)	31	54.0 (21.0, 20–100)
On ASM or benzodiazepine treatment	33	6 (9.1)

Abbreviations: SD = Standard Deviation, FDS = Functional/dissociative seizures, ASM = Antiseizure medication, HQoL-VAS = Health-related Quality of Life, Visual Analogue Scale.

Table 2

Descriptive statistics for full-scale IQ, for tests of executive function and for self-reported executive function based on the Behavior Rating Inventory of Executive Functioning in patients with functional/dissociative seizures assessed at baseline.

	N	Minimum	Maximum	Mean (SD)	Dysfunction n (%)
Estimated IQ	33	57	123	96.9 (14.9)	2 (6.1)
Letter-number span	33	27	63	43.9 (9.2)	4 (12.1)
T-score				43.2 (11.3)	7 (21.2)
TMT-B T-score	30	6	61	56.1 (12.0)	9 (27.3)
Stroop-Color-Word	33	37	67	45.1 (11.1)	7 (21.2)
T-score				58.8 (10.6)	8 (24.2)
EF composite T-score	33	22	66	58.1 (11.5)	9 (27.3)
BRIEF behavioral regulation index T-score	32	36	77	56.1 (12.0)	10 (30.3)
BRIEF metacognition index T-score	32	37	67		
BRIEF global executive composite T-score					

Abbreviations: TMT = Trail Making Test, BRIEF = Behavior Rating Inventory of Executive Functioning.

Table 3

Baseline data by FDS outcome Participants with IQ < 70 were excluded ($n = 2$). Cutoff: IQ ≤ 78 , T-score ≤ 35 , BRIEF ≥ 65 .

Variables	FDS-free at follow up (n = 14)	Continued to experience FDS at follow up (n = 19)	P value
Female sex, n (%)	10 (71.4)	18 (94.7)	0.07
Age at inclusion, mean (SD, range)	30.1 (12.7, 17–52)	35.0 (14.6, 16–62)	0.33
Age at FS onset, mean (SD, range)	22.6 (10.0, 9–43)	29.8 (13.7, 8–56)	0.15
Diagnostic delay in years, mean (SD, range)	6.7 (8.2, 0.3–25.3)	4.3 (7.4, 0.2–30.9)	0.37
FS frequency at baseline, monthly, mean (SD, range)	14.2 (17.2, 1–50)	13.1 (17.0, 0.2–70)	0.86
Education in years, mean (SD, range) (n = 31)	13.5 (3.6, 9–22)	13.1 (2.2, 10–17)	0.70
Marital status, married/partner, n (%)	7 (50)	9 (47.3)	0.88
Employment, employed/student, n (%)	7 (50)	5 (26.3)	0.16
Psychiatric history for anxiety or depression, n (%) (n = 30)	4 (33.3)	11 (61.2)	0.14
Psychotherapy prior to inclusion, n (%) (n = 31)	8 (66.7)	11 (57.9)	0.63
HQoL-VAS, mean (SD, range) (n = 31)	60.0 (22.2, 30–100)	50.3 (20.0, 20–95)	0.12
On ASM or benzodiazepine treatment, n (%)	2 (14.3)	1 (5.2)	0.23
Full-scale IQ below cutoff, n (%)	0 (0)	2 (10.5)	0.21
Executive score below cutoff, n (%)	2 (14.3)	5 (26.3)	0.40
BRIEF Behavioral Regulation Index above cutoff, n (%)	2 (14.3)	6 (31.6)	0.25
BRIEF Metacognitive index above cutoff, n (%) (n = 32)	1 (7.1)	8 (44.4)	0.02*

Abbreviations: SD = Standard Deviation, FDS = Functional/dissociative seizures, QoL-VAS = Health-related Quality of Life, Visual Analogue Scale, ASM = Antiseizure medication, IQ = Intelligence Quotient, BRIEF = Behavior Rating Inventory of Executive Functioning.

Table 4

High Metacognitive Index score (≥ 65) versus normal Metacognitive Index score (< 65) (chi-square and independent t test, $n = 33$).

Variables	High score on metacognitive index (n = 10)	Normal score on metacognitive index (n = 23)	P value
Female sex, n (%)	10 (100)	18 (78.3)	0.11
Age at inclusion, mean (SD, range)	34.1 (16.6, 17–62)	32.7 (12.4, 17–57)	0.81
Age at FDS onset, mean (SD, range)	28 (14.7, 12–56)	26.0 (12.0, 8–49)	0.72
Diagnostic delay in years, mean (SD, range)	3.8 (4.9, 0.2–16.3)	5.9 (8.7, 0.3–31.0)	0.39
FS frequency at baseline, monthly mean (SD, range)	12 (12.1, 1–30)	14.3 (18.7, 0.2–70)	0.68
Education in years, mean (SD, range)	12.4 (2.3, 10–17)	13.6 (3.1, 9–22)	0.49
Marital status, married/partner, n (%)	5 (50)	10 (43.5)	0.73
Employment, employed/student, n (%) (n = 32)	2 (22.2)	9 (39.1)	0.37
Psychiatric history of anxiety or depression, n (%) (n = 31)	8 (88.9)	8 (38.1)	0.01*
Psychotherapy prior to inclusion, n (%) (n = 31)	7 (70)	12 (57.1)	0.49
HQoL-VAS, mean (SD, range)	43.5 (18.3, 20–70)	59.1 (20.8, 25–100)	0.05*
On ASM or benzodiazepine treatment, n (%)	0 (0)	3 (13)	0.20
Full-scale IQ below cutoff, n (%)	1 (10)	0 (0)	0.12
Executive score below cutoff, n (%)	4 (44.4)	3 (13.0)	0.053
BRIEF Behavioral Regulation Index above cutoff, n (%)	6 (60)	3 (13.0)	0.005*
FS-free at follow-up, n (%)	1 (11.1)	13 (56.5)	0.02*
FS reduction $\geq 50\%$ at follow-up, n (%)	5 (55.6)	22 (95.7)	0.005*

Abbreviations: SD = Standard Deviation, FDS = Functional/ dissociative seizures, ASM = Antiseizure medication, HQoL-VAS = Health related Quality of Life Visual Analogue Scale, BRIEF = Behavior Rating Inventory of Executive Functioning.

Table 5

Significant predictors of seizure free outcome at follow-up in patients with FDS (multivariate logistic regression analysis).

	Odds ratio (CI)	P
Age at FDS onset	1.06 (0.99 – 1.14)	0.11
Metacognitive index below cutoff	10.47 (1.04 – 106.02)	0.047*

* $p \leq 0.05$, Nagelkerke $R^2 = 0.32$. Abbreviations: FDS = Functional dissociative seizures, CI = Confidence Interval.

dysfunction measured by the MI from BRIEF was an independent risk factor for ongoing FDS at a mean of 5.5 years of follow-up. In contrast, objective measures of EF showed no significant difference between patients who had become FDS free at follow-up and those who had not. Furthermore we found that self-reported executive dysfunction was associated with a history of anxiety and depression, and reduced health related quality of life

4.1. Subjective and objective measures of executive functioning

That self-reported executive dysfunction measured by the MI from

BRIEF was an independent risk factor for ongoing FDS at follow-up appears plausible: EF encompasses multiple domains, such as planning, inhibition, set shifting, self-monitoring, organization, and initiating and sustaining mental activity, and is crucial for goal achievement. The MI score reflects ability to solve problems and to monitor success and failure; both these traits are important for achieving a therapeutic goal and for following, or adapting and then following, a therapeutic plan. Therefore, it is possible that a high MI score is associated with lower likelihood of successful therapy. Indeed, it has been shown that EF is associated with treatment retention in other patient populations [26]. In contrast, objective measures of EF showed no significant difference between patients who had become FDS free at follow-up and those who had not. A lack of correlation between self-reported EF and performance on tests for EF has previously been described for patients with epilepsy [25] or for other neurological disorders [26]. It has been argued that complex functions, such as EF, may be challenging to assess by structured neuropsychological tests, as they resemble the complexities of everyday life situations only to a limited extent. Neuropsychological tests may therefore be of restricted ecological validity when assessing EF [27]. This previously reported weak correlation between tested and self-reported EF [28,29], has led to the conclusion that tests and self-reports reflect different aspects of EF. Taken together it can be argued that both subjective and objective measures of EFs may be useful to identify individuals that could benefit of specifically adjusted treatment programs and or cognitive rehabilitation.

Discrepancies between subjective and objective measures, not of cognitive function, but of distress, arousal, and symptom burden in patients with FDS, have been reported previously. One meta-analysis found no correlation between subjective and objective measures in most of the included studies, raising the question of whether subjective or objective measures are more meaningful [30]. The authors concluded that subjective and objective measures assess different, but equally valid, constructs and argue that both might be meaningful in understanding and exploring different aspects of FDS further [30]. Impaired EF as assessed by neuropsychological examination in patients with FS has been previously described; one study showed reduced performance on the Trail Making Test and Digit Span Test Forward and Backward in 40 patients with FDS, indicating lower working memory and set-shifting ability, compared with healthy controls [31,32,33]. Taken together this emphasizes the importance of taking both test results and self-report into account when conducting an evaluation of overall functioning.

4.2. Executive functioning, emotional distress and quality of life

Self-reported executive dysfunction was also associated with a history of anxiety and depression in our study. There is evidence that mood disorders are frequently accompanied by a range of cognitive deficits [34,35]. Deficits in EF have been linked to depressive states, both active [36] and in remission [37], and also to anxiety disorders [38]. Taken together this may indicate that executive dysfunction and emotional distress may have a mutual effect on each other; emotional distress may reduce cognitive/executive capacity, and executive dysfunction may result in maladaptive functioning leading to emotional distress.

We also found an association between a high (abnormal) score on the MI of the BRIEF and health-related quality of life. These findings are consistent with previous research demonstrating that executive dysfunction as measured with BRIEF is related to poor quality of life [39] in other patients populations, including in children with epilepsy [40] and adults with ADHD [41]. Executive dysfunction is related to social dysfunction, and reduced educational [42] and occupational [43] attainment, and is therefore likely to impact health-related quality of life.

In summary, screening of EF by a self-report questionnaire could easily be incorporated into the clinical assessment of patients with FDS. Results could provide useful prognostic information, and help tailoring psychological interventions to individual patients.

The study's primary strengths lie in its extended follow-up period, minimal participant attrition, exclusive focus on patients with documented functional neurological disorder (FND) according to ILAE guidelines, and comprehensive neuropsychological assessments. Our study does, however, have limitations. These are mainly the small sample size and recruitment from a specialized epilepsy centre, which increases vulnerability to selection bias and thereby limits the generalizability of our findings. The fact that we did not carry out a power analysis is also a significant limitation as we cannot demonstrate that our study was adequately powered. There is also a selection bias from the fact that people were recruited for an inpatient hospital stay. In addition, the wide difference, of up to ten years, between recruitment and follow up in individual patients may have affected our results.

5. Conclusion

In conclusion, our results showed that among patients with FDS, self-reported problems with EF were associated with long-term outcome. Whether this is due to a direct relationship with FDS chronicity, a limitation regarding the effectiveness of therapy, or both is not clear. Screening of EF using a self-report questionnaire could easily be incorporated into the clinical assessment of appropriate patients with FDS. The results could provide useful prognostic information, as well as potentially contributing towards tailoring psychological interventions to individual patients.

Contributions of authors' statement

All co-authors have been substantially involved in the study and/or the preparation of the manuscript. The corresponding author attests that all listed authors meet authorship criteria and that there are no undisclosed groups or persons who have had a primary role in the study or in manuscript preparation. All co-authors have seen and approved the submitted version of the paper and accept responsibility for its content.

Declaration of competing interest

A.V. reports personal fees from Eisai and Arvelle Therapeutics, unrelated to this study. R.D. receives royalties from UpToDate. M. I. Lossius reports personal fees from Eisai, UCB, and Arvelle, outside the submitted work. The remaining authors have no conflicts of interest. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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